Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/30/2014
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Investigation of Host-Pathogen Interactions, Conserved Cellular Responses, and Countermeasure Efficacy During Spaceflight using the Human Surrogate Model Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Immunology		
Space Biology Special Category:	None		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/13/2010	**End Date:**	02/29/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 2/29/2016 per NSSC information (Ed., 2/25/15) NOTE: End date is now 2/28/2015 per NSSC information (Ed., 3/13/14) NOTE: End date is now 7/12/2014 per NSSC information (Ed., 2/1/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alverdy, John (University of Chicago) Ott, C Mark (NASA Johnson Space Center)		
Grant/Contract No.:	NNX10AO52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Changes that occur to both the host immune system and pathogenesis of microbes during spaceflight represent a formidable challenge to the successful transition from short-to-long-duration spaceflight. This is a critical concern, since a) the immune system of the crew is dysfunctional during spaceflight, b) spaceflight uniquely increases the virulence and globally alters gene expression of the human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect ground-based animal models, c) spaceflight-induced increases in S. typhimurium virulence are regulated by media ion composition, d) phosphate ion is sufficient to alter related pathogenesis responses in a ground-based spaceflight analogue model, and e) the evolutionarily conserved bacterial RNA chaperone protein, Hfq, serves as a master molecular regulator of many of these responses. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight - none of these studies have been done when both the host and pathogen are simultaneously exposed to the spaceflight environment. To address this knowledge gap, we propose to challenge the human surrogate model Caenorhabditis elegans with S. typhimurium during spaceflight to study the role of the intestine in host-pathogen interactions in real-time, define virulence mechanisms, identify evolutionarily conserved responses, and test novel therapeutic strategies to prevent infectious disease. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated. We further propose that these responses represent evolutionarily conserved mechanisms at the level of both host and pathogen that are dependent on media ion composition, specifically the local concentration of phosphate. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Previously published results from our lab showed that spaceflight uniquely increases the virulence and globally alters gene expression of the major human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect laboratory-based animal models. In our efforts to determine the mechanism(s) behind the spaceflight-induced responses in Salmonella, we discovered that a) spaceflight-induced increases in Salmonella virulence are regulated by media ion composition, b) phosphate ion is sufficient to alter related pathogenesis responses in a spaceflight analogue model, c) the Hfq protein serves as a master molecular regulator of many of these responses in Salmonella, and d) Hfq also acts as a spaceflight-induced regulator acting across bacterial species, which to our knowledge, is the first report of such a master response regulator of the spaceflight-induced stimulon in bacteria. Hfq is a highly conserved bacterial RNA chaperone protein that binds to small non-coding regulatory RNAs thereby facilitating their association with mRNAs, the result of which plays a diverse role in prokaryotic gene expression (including regulation of ion response pathways), virulence, and physiology in response to stress. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight – no studies have been published that document the outcome when both the host and pathogen are simultaneously exposed to the spaceflight environment. Furthermore, the eukaryotic organism and human surrogate infection model, C. elegans, contains structural and functional homologues of Hfq proteins, which are a family of evolutionarily conserved RNA-binding proteins that are critical for immune function, cell differentiation/function, and responses to cellular stress. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated and may uncover common conserved responses in the host and pathogen that can be targeted for countermeasure development. We have made significant progress toward achieving the objectives to successfully support a spaceflight investigation to meet a current SpaceX-5 launch date of Dec 3, 2014, including: - Finalized all experimental parameters for successful completion of our experiment (loading volumes, timing, etc.). - Worked closely with Bioserve to finalize hardware design and development to support this work on orbit. This includes a successful modification of video capabilities on orbit in order to provide required resolution of infected nematodes. - Projected successful completion of our EVT in early July. EVT operations were initiated in early June and all operations to this point have gone smoothly and successfully. Video analysis and RNA integrity assessment will be completed in early July. - We are on target for the projected Micro-5 launch date of December 3, 2014.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Crabbe A, Nielsen-Preiss S, Woolley CM, Barrila J, Buchanan K, McCracken J, Inglis DO, Searles SC, Nelman-Gonzalez MA, Ott CM, Wilson JW, Pierson DL, Stefanyshyn-Piper HM, Human LE, Nickerson CA. "Spaceflight enhances cell aggregation and random budding in Candida albicans." PLoS One. 2013;8(12): e80677. Published 04 Dec 2013. http://dx.doi.org/10.1371/journal.pone.0080677 ; PMID: 24324620 , Dec-2013
Articles in Peer-reviewed Journals	Crabbé A, Ledesma MA, Nickerson CA. "Mimicking the host and its microenvironment in vitro for studying mucosal infections by Pseudomonas aeruginosa." Pathogens and Disease. 2014 Jun;71(1):1-19. http://dx.doi.org/10.1111/2049-632X.12180 ; PubMed PMID: 24737619 , Jun-2014
Articles in Peer-reviewed Journals	Kennedy AR, Crucian B, Huff JL, Klein SL, Morens D, Murasko D, Nickerson CA, Sonnenfeld G. "Effects of sex and gender on adaptation to space: immune system." Journal of Women's Health. 2014 Nov;23(11):956-8. http://dx.doi.org/10.1089/jwh.2014.4913 ; PubMed PMID: 25401940 ; PubMed Central PMCID: PMC4236064 (NOTE: reported originally as "Accepted as of July 2014") , Nov-2014
Awards	Nickerson C. " Editor-in-Chief of a new Nature Publishing Group journal, npj:Microgravity, dedicated to microgravity research - journal launch, Fall 2014." Jul-2014
Awards	Nickerson C. "Invited Committee Member and Session Chair, Institute of Medicine, National Academy of Sciences review of NASA's Evidence Reports of Human Health Risks, July 2014." Jul-2014
Significant Media Coverage	Williams R. "Spaceflight research highlighted in a radio interview by Robyn Williams for the Australian radio program, The Science Show. Session title 'Experiments in space reveal genetic master switches in pathogens and vaccines.' " Australian radio program, The Science Show. March 29, 2014. http://www.abc.net.au/radionational/programs/scienceshow/experiments-in-space-reveal-genetic-master-switches-in-pathogen/5352724 , Mar-2014
Significant Media Coverage	Boeing video. "ISS Discovery - A more potent disease in space. A Boeing featured video on value of using the microgravity research platform for innovations in infectious disease research." Boeing featured video, January 27, 2014. http://youtu.be/GmG9pfc6E74 , Jan-2014

Significant Media Coverage	O'Brien M. "Spaceflight research highlighted on the PBS News Hour television documentary series with science correspondent Miles O’Brien. Series will air in 2014." PBS News Hour television documentary series with science correspondent Miles O’Brien, 2014., Jul-2014